Yuchen Liu

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AI systems engineer building RL training infrastructure, search/data pipelines, and research-grade open-source tools.

EDUCATION

M.S. Computer Science	University of Pennsylvania, Philadelphia	GPA: 3.80/4.00	May 2025
B.S. Applied Economics	Pennsylvania State University, State College	GPA: 3.86/4.00	May 2023

WORK EXPERIENCE

Software Engineer, Search Infrastructure | DoorDash — Aug 2025 - Present | Seattle, WA

- Built **Notus**, a next-generation search ingestion platform using **Iceberg, Kafka, and Spark** to decouple document ingestion from serving, enable configuration-driven indexing, and unblock schema evolution for hybrid search.
- Built **Vela**, Argo's hybrid retrieval platform, enabling **Lucene-native hybrid search** and federated retrieval with vector backends such as **Milvus** behind a single API.
- Migrated **30+ production stacks** to modular **Jsonnet**, eliminating roughly **56,000 lines** of duplicate configuration and reducing config verification time by **90%**.
- Stabilized production indexers with **500GB EBS storage**, pre-deployment validators, and **Kyverno guardrails** after RCA for **$21K revenue-risk incidents**, preventing capacity-related outages.

Software Engineering Intern | Penn Medicine TissueLab — Oct 2024 - May 2025 | Philadelphia, PA

- Developed a **Python/FastAPI** AI microservice platform to orchestrate **PyTorch** models for medical image analysis, enabling natural-language-driven segmentation and classification workflows.
- Implemented an event-driven **DAG workflow engine** for async task execution, processing **10,000+ pathology images daily** and streaming **12M+ imaging events** through WebSockets with sub-10ms latency.
- Integrated generative and discriminative deep learning models into a unified service layer with **asynchronous REST APIs** for cross-department usage.

Software Engineering Intern | Information Technology of CAS — Apr 2024 - Aug 2024 | Remote

- Designed **Kafka and Redis-backed** event streaming services that reduced **p95 API latency from 2s to 200ms** while processing millions of IoT events.
- Contributed to distributed **Spring Boot** monitoring services, delivered **35+ REST APIs**, and maintained **99.7% uptime** in Unix-based production environments.

OPEN SOURCE

nanoRL

- Led an open-source **RL training framework** inside nanochat for clean objective definitions, rollout experiments, RL algorithm research, reproducible ablations, and **vLLM inference serving**.

nanochat

- Built a hackable pretraining and chat stack supporting architecture definition and **FLOP-controlled model ablations**.

llm-interp

- Authored reproducible interpretability scripts for model circuit research, including **attention-sink analysis** and reproductions of **LLM decode indeterminism** findings.

RESEARCH

Do Value Vectors in Deep Layers Need Context from the Residual Stream? — EMNLP 2026, under review

Co-first author. Challenged the standard attention mechanism of computing value vectors from the residual stream, finding that deep layers benefit from **context-free value vectors** that preserve original token information.

Proposed **Bank of Values**, a learned value-vector table for the last third of layers that eliminates the **V cache** and improves validation loss and average scores across **21 benchmarks** on **135M and 780M** models.

Vision Language Models Cannot Plan, but Can They Formalize? — ECCV 2026, under review

Studied whether **VLMs** can formalize visual planning tasks into solver-executable **PDDL problem files**.

A co-evolving agentic AI system for medical imaging analysis — Nature, under review

Built agentic medical-imaging workflows for **tool selection, planning, knowledge updates**, and human-in-the-loop co-evolution.

SKILLS

Languages: Python, Kotlin, Java, C++, JavaScript; ML/Infra: PyTorch, vLLM, Spark, Kafka, Iceberg, Delta Lake, FastAPI, Redis, Docker, Kubernetes, AWS, Spring Boot